MIL-T-1180D
5 August 1981
SUPERSEDING
MIL-T-1180C(ME)
21 April 1976

MILITARY SPECIFICATION

THERMOMETER, SELF-INDICATING, LIQUID-IN-GLASS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a liquid-in-glass, self-indicating thermometer graduated in Farenheit (F) degrees, for indicating temperatures of photographic solutions.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The thermometer shall be liquid-in-glass, glass tube with the bulb at one end, graduated in Fahrenheit degrees. The thermometer shall be mounted on a metal back.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3 Storage conditions. The thermometer shall not be damaged by storage in any ambient temperature from minus 28° F to plus 145° F (minus 33° C to plus 63° C).

3.4 Thermometer tube.

3.4.1 Stem. The stem shall consist of a sealed magnifying-type glass tube of uniform bore, with a bulb at one end, and shall have a broad stripe of white glass at the back to increase reflected light. The stem shall be designed to increase the angle of vision and effective width of the thermo-sensitive fluid column. The stem shall be not less than 5 inches nor more than 5.25 inches long, and shall be 0.266 inch in diameter.

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3.4.2 Thermosensitive fluid. The bulb and tube shall contain a sufficient quantity of thermosensitive fluid so that, when correctly calibrated, the height of the fluid in the tube shall indicate the temperature of the medium surrounding the thermometer within the range and accuracy specified herein. The type of fluid used shall be such so as to minimize any tendency for the liquid column to separate. A non-fading dye may be used to color the fluid to allow a contrast for reading. The color of the fluid shall permit reading the thermometer under average darkroom conditions. The thermometer shall show a change in temperature within 30 seconds when transferred from water at a temperature of plus 60° F (plus 16° C) to water at a temperature of plus 90° F (plus 32° C).

3.4.3 Temperature range. The thermometer shall have a total temperature range of not less than plus 40° F to plus 140° F. Within the range of plus 60° F to plus 90° F, the thermometer shall indicate any temperature change within plus or minus 0.5° F in a period of 30 seconds when totally immersed. Within the range of plus 50° F to plus 60° F and plus 90° F to plus 140° F, the thermometer shall have an accuracy of plus or minus 1° F when totally immersed.

3.4.4 Scale. Scale markings shall be legibly etched or fused on the thermometer stem. No scale marking shall be on the metal back plate. An arrow or heavy index mark shall be used to indicate the plus 68° F mark. Graduation marks shall be every 2 degrees, with every 10 degrees numbered. Graduation marks shall be clean cut, straight, and of uniform width. The width shall not exceed one-fifth the interval between graduations, and the separation between graduations shall be not less than 0.0625 inch. Graduation marks shall be clearly visible under normal darkroom conditions.

3.5 Metal back. The thermometer shall be mounted on a 22 gage (Birmingham gage 0.031 inch) thick corrosion-resisting steel back plate. The back plate shall extend at least the full length of the thermometer and its bulb, and shall be curved to provide protection for the thermometer. The back shall have a hook of the same material as the back plate permanently attached at the top. The hook shall be designed so that, when the thermometer is used in a developing tray, the hook will act as a standard for tilting the thermometer to a convenient angle for reading. When used in a developing tank, the hook shall enable the thermometer to hang freely from the edge of the tank.

3.6 Chemical resistance. The thermometer stem, the markings on the tube, the back plate, the hook, and the fasteners shall not be damaged when tested as specified in 4.4.2.3.

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3.7 Identification marking. The thermometer shall be identified in accordance with MIL-STD-130.

3.8 Workmanship. The thermometer and thermosensitive fluid shall be free from gas bubbles and foreign material. The thermometer shall be clean and free from pits, chips, sharp edges, or other defects which would prove detrimental to accuracy and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defects.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

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4.4 Inspection procedure.

4.4.1 Examination. The thermometer shall be examined as specified herein for the following defects:

Major

101. Thermometer stem not as specified.
102. Thermosensitive fluid not as specified.
103. Scale markings not as specified.
104. Arrow or index mark indicating 68° F missing or not in correct position.
105. Back plate not as specified.
106. Hook missing or not as specified.
107. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

4.4.2 Tests.

4.4.2.1 Test conditions. A thermometer which has been certified by the National Bureau of Standards to have an accuracy of plus or minus 0.1° F shall be used in the tests specified in 4.4.2.1.2 through 4.4.2.1.4.

4.4.2.1.1 Storage tests. Sample thermometers shall be placed in a cold chamber at a temperature of minus 28° F (minus 33° C) for not less than 6 hours. The thermometers shall then be removed, allowed to return to room temperature, and placed in an oven at plus 145° F (plus 63° C) for 6 hours. The thermometers shall be removed from the oven, allowed to return to room temperature, and examined. Any evidence of breaks, cracks, or separation bubbles in the thermosensitive fluid shall constitute failure of this test.

4.4.2.1.2 Temperature range (60° F to 90° F). The certified thermometer and the sample thermometers shall be immersed in a water solution at a temperature of plus 60° F and the readings of the certified and sample thermometers compared. The temperature of the solution shall be raised to plus 90° F and readings taken every 6° F. Failure of the sample thermometers to indicate the same reading as the certified thermometer within 0.5° F in 30 seconds shall constitute failure of this test.

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4.4.2.1.3 Temperature range (50° F to 60° F). The sample thermometers shall be tested as specified in 4.4.2.1.2, except that the temperature of the solution shall be raised from plus 50° F to plus 60° F and readings taken every 2° F. Failure of the sample thermometers to indicate the same reading as the certified thermometer within plus or minus 1.0° F shall constitute failure of this test.

4.4.2.1.4 Temperature range (90° F to 140° F). The sample thermometers shall be tested as specified in 4.4.2.1.2, except that the temmperature of the solution shall be raised from plus 90° F to plus 140° F and readings taken every 10° F. Failure of the sample thermometers to indicate the same temperature as the certified thermometer within plus or minus 1.0° F shall constitute failure of this test.

4.4.2.2 Indication speed. The thermometers shall be tested for speed indication by total immersion in a water solution at plus 60° F until stabilized, and then totally immersed in a water bath at 90° F. After stabilized at 90° F, the thermometers shall be transferred to the 60° F water solution. Failure to indicate a temperature change within 30 seconds shall constitute failure of this test.

4.4.2.3 Chemical resistance. The thermometers shall be totally immersed in a 5-percent solution of sodium carbonate (NA_2CO_3) for not less than 12 hours. The thermometers shall be rinsed in water and totally immersed in a 5-percent solution of sulfuric acid (H_2SO_4) for not less than 12 hours. The thermometers shall be rinsed in water, dried, and examined. Any evidence of damage, breakage, or evidence of chemical action to the thermometer, scale marking, back plate, hook, or fasteners shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Methods and containers not as specified for Level A or B. Each incorrect method or container shall constitute one defect.
109.	Cushioning not provided as specified.
110.	Quantity placed in consolidated container not as specified for Level A.

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111. Boxes not closed as specified for Level A or B.
112. Commercial packaging not in accordance with referenced documents.
113. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each thermometer shall be cushioned to prevent damage to the graduated surface and back plate and shall be preserved in accordance with MIL-P-116, Method III.

5.1.1.1 Consolidation. The thermometers shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, style optional in quantities as specified (see 6.2). Box closure shall be in accordance with Method V of the appendix to the box specification.

5.1.2 Commercial. Thermometers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Thermometers preserved as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Thermometers, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style optional. The boxes shall be closed in accordance with Method V of the appendix to the box specification.

5.2.3 Commercial. Thermometers shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (Level A or Level B), shall be in accordance with MIL-STD-129.

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5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The thermometer is intended for use in maintaining photographic processing chemicals in tanks or trays at the correct temperature.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(c) Quantities to be consolidated together (see 5.1.1.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
Army - ME

Review activities:
Army - ER
DLA - GS

User activity:
Air Force - 99

Preparing activity:
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DOCUMENT IDENTIFIER (Number) AND TITLE
MIL-T-1180D Thermometer, Self-Indicating, Liquid-In-Glass

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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